

ECHS MATHEMATICS

IB Diploma Programme - Mathematics: Applications and Interpretation SL

1.1

Scientific Notation, Approximation and Error

Full Notes - Scale - Precision - Bounds - Error - TI-84 Plus CE

Central idea. A calculator result becomes useful mathematics only when its **scale, units, precision**, and **uncertainty** are communicated clearly.

Represent → Calculate → Estimate → Bound → Interpret



Name: _____ Class: _____

Date: _____ Version: Student Full Notes

Learning route and scope

current IB SL core first - optional prerequisites separated

Learning intentions

By the end of Lesson 1.1, you should be able to:

1. convert between ordinary notation and normalized scientific notation;
2. multiply, divide, add, and subtract quantities written with powers of ten;
3. compare scale using exponents, ratios, and estimates;
4. convert length, area, and volume units without losing the correct power;
5. use TI-84 Plus CE EE entry, SCI/NORM display, brackets, and guard digits transparently;
6. round correctly to decimal places and significant figures while preserving meaningful zeros;
7. distinguish exact values, approximations, and estimates;
8. construct direct and calculated bounds;
9. calculate absolute, relative, and percentage error and interpret tolerance;
10. communicate precision and uncertainty honestly in an IB-style conclusion.

Core reference sheet

Normalized scientific notation

$$N = a \times 10^k, \text{ where } 1 \leq |a| < 10 \text{ and } k \in \mathbb{Z}.$$

Multiply

$$(a \times 10^m)(b \times 10^n) = ab \times 10^{m+n}, \text{ then normalize.}$$

Divide

$$\frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n}, \text{ then normalize.}$$

Add/subtract

Rewrite with a common power of ten before combining coefficients.

Error interval

$$\text{If } x_0 \text{ is rounded to nearest unit } h: x_0 - \frac{h}{2} \leq x < x_0 + \frac{h}{2}.$$

Absolute error

$$E_a = |A - X|.$$

Relative error

$$E_r = \frac{|A - X|}{|X|}.$$

Percentage error

$$E_{\%} = \frac{|A - X|}{|X|} \times 100\%.$$

Guard digits

Keep full calculator precision through working; round once at the end.

Technology principle

GDC SKILL The TI-84 is used here to enter scientific notation efficiently, change display format, and preserve guard digits. **GDC NOT NEEDED** Normalization, exponent structure, bounds, and simple rounding should remain mathematically visible. **GDC EXPECTED** In multi-step contextual calculations, calculator use is appropriate but must be estimated and interpreted.

IB communication tip

A strong response does not copy an E-display blindly. Rewrite the result using $\times 10^k$, preserve justified significant figures, attach units, and check whether the scale is reasonable.

1.1A - Scientific Notation and Scale

normalized form, operations, units and reasonableness

The normalized form

A non-zero number is in normalized scientific notation when

$$N = a \times 10^k, \quad 1 \leq |a| < 10, \quad k \in \mathbb{Z}.$$

The coefficient a carries the significant digits; the exponent k communicates scale.

Worked Example 1

GDC NOT NEEDED

Write 72 900 000 in normalized scientific notation.

WORKING / ANNOTATIONS

Your Turn 1

GDC NOT NEEDED

Write each quantity in normalized scientific notation.

- (a) 0.000438
- (b) 905 000 000 000
- (c) -0.00000604

WORKING / ANNOTATIONS

Moving back to ordinary notation

A positive exponent moves the decimal to the right when expanding. A negative exponent moves it to the left.

$$8.021 \times 10^6 = 8\,021\,000, \quad 5.7 \times 10^{-3} = 0.0057.$$

The sign of the exponent describes *scale*; it does not determine whether the number itself is positive or negative.

Worked Example 2

GDC NOT NEEDED

Calculate $(4.8 \times 10^7)(2.5 \times 10^{-3})$.

WORKING / ANNOTATIONS

Your Turn 2

GDC NOT NEEDED

Calculate and normalize:

$$\frac{9.6 \times 10^{-4}}{3.2 \times 10^2}.$$

WORKING / ANNOTATIONS

Addition and subtraction

The exponent law does **not** allow powers to be added directly. First use a common power of ten:

$$4.2 \times 10^5 + 7.0 \times 10^4 = (4.2 + 0.70) \times 10^5 = 4.90 \times 10^5.$$

For subtraction:

$$5.1 \times 10^8 - 7.6 \times 10^7 = (5.1 - 0.76) \times 10^8 = 4.34 \times 10^8.$$

Common trap

The statement $3.4 \times 10^5 + 2.1 \times 10^4 = 5.5 \times 10^9$ is structurally impossible. Before adding, align powers: $(3.4 + 0.21) \times 10^5 = 3.61 \times 10^5$.

Compare scale before coefficients

For positive normalized quantities, compare exponents first. If exponents match, compare coefficients.

$$7.9 \times 10^6 > 1.2 \times 10^6 > 8.2 \times 10^5.$$

A difference of 5 powers of ten represents a scale factor of approximately 10^5 .

Unit conversions - the power matters

If $1 \text{ km} = 10^3 \text{ m}$, then

$$1 \text{ km}^2 = 10^6 \text{ m}^2, \quad 1 \text{ km}^3 = 10^9 \text{ m}^3.$$

Area and volume conversions require squaring or cubing the *conversion factor*, not merely changing the unit label.

Worked Example 3**GDC NOT NEEDED**

Convert $4.82 \times 10^{-6} \text{ km}^2$ to square metres.

WORKING / ANNOTATIONS**Your Turn 3****GDC NOT NEEDED**

A cube has edge $3.2 \times 10^{-3} \text{ m}$. Find its volume in m^3 .

WORKING / ANNOTATIONS**Reasonableness estimate**

Before pressing ENTER, round coefficients sensibly and use exponent structure. For example,

$$\frac{6.2 \times 10^7}{3.1 \times 10^3} \approx \frac{6 \times 10^7}{3 \times 10^3} = 2 \times 10^4.$$

The exact quotient is also 2×10^4 . A result near 10^{11} would signal an entry-order error.

Checkpoint - 1.1A

1. Write 0.0000000817 in normalized scientific notation.
2. Calculate $(6.0 \times 10^{-7})(8.0 \times 10^{11})$.
3. Calculate $6.2 \times 10^{-3} - 7.5 \times 10^{-4}$.
4. Convert 7.5×10^8 mm to km.
5. Explain why 45.2×10^3 is not normalized.

WORKING / ANNOTATIONS**1.1B - TI-84 Plus CE and Validation**

EE entry, SCI/NORM, brackets and guard digits

TI-84 workflow A1 - EE entry**GDC SKILL** Enter 4.20×10^{-4} while preserving the intended precision.

1. Type **4 . 2 0**.
2. Press **2nd** then **,** for **EE**.
3. Press the dedicated **(-)** key, then **4**.
4. Press **ENTER**.

The display may show 4.2E-4; write the final mathematical value as 4.20×10^{-4} if three significant figures are required.

Common trap

The TI-84 negative key **(-)** and subtraction key are not the same operation. In an exponent such as 10^{-4} , use the negation key.

TI-84 workflow A2 - SCI and NORM

GDC SKILL To display 0.00000672 in scientific notation:

1. **MODE** → highlight **SCI** → **ENTER**.
 2. Quit the mode screen, then evaluate 0.00000672.
 3. Read 6.72E-6 as 6.72×10^{-6} .
 4. Restore **NORMAL** when the demonstration is complete.
- SCI/NORM changes the display, not the stored numerical value.

TI-84 workflow B1 - brackets and guard digits

GDC EXPECTED Evaluate

$$\frac{4.73 \times 10^8}{6.2 \times 10^{-3}}$$

Enter the numerator and denominator as complete bracketed scientific-notation numbers. The calculator gives

$$7.629032258 \dots \times 10^{10}$$

Retain guard digits and round only the final answer:

$$7.6 \times 10^{10} \text{ to 2 s.f.}$$

The one-significant-figure estimate $\frac{5 \times 10^8}{6 \times 10^{-3}} \approx 8 \times 10^{10}$ confirms the scale.

Your Turn 4

GDC EXPECTED

A satellite stores 18 images. Each image has 2.40×10^8 pixels, each pixel uses 3.20×10^1 bits, and 8 bits = 1 byte. Find the total number of bytes. Keep guard digits until the end.

WORKING / ANNOTATIONS

Technology evidence checklist

Before accepting calculator output, ask:

- Did I enter the intended grouping and exponent signs?
- Does the exponent match a mental estimate?
- Are the units correct after multiplication/division/conversion?
- Have I kept guard digits until the final reporting step?
- Have I rewritten E notation as standard mathematical notation?

1.1C - Precision, Rounding and Approximation

decimal places, significant figures and honest reporting

Decimal places versus significant figures

Decimal places (d.p.) count positions to the right of the decimal point. **Significant figures (s.f.)** begin at the first non-zero digit and communicate measurement precision.

$$47.38621 \text{ to 3 d.p.} = 47.386, \quad 0.004986 \text{ to 3 s.f.} = 0.00499.$$

Worked Example 4

GDC NOT NEEDED

Round 38.7462 to 2 decimal places and -12.499 to 1 decimal place.

WORKING / ANNOTATIONS

Significant zeros

Zeros can be meaningful.

- Leading zeros in 0.00420 are not significant; the final zero is significant: 3 s.f.
- Interior zeros in 3.0400 are significant: the number has 5 s.f.
- 72000 is ambiguous; 7.20×10^4 communicates exactly 3 s.f.

Your Turn 5

GDC NOT NEEDED

- Round 58492 to 3 s.f.
- Round 83.596 to 2 d.p.
- Write 72000 unambiguously to 3 s.f.
- State the number of significant figures in 6.040×10^{-7} .

WORKING / ANNOTATIONS

Exact value, approximation, estimate

Exact value Defined without rounding, e.g. $\frac{1}{3}$, $\sqrt{2}$, π .

Approximation A nearby reported value, e.g. $\pi \approx 3.142$.

Estimate A deliberately simplified value used to predict scale, e.g. $49.8(201) \approx 50(200)$.

Use $=$ for equality and $\approx$ for approximation.

Common trap

Premature rounding can move the final result. Keep the original measurement and full calculator value through working. Round early only when you are deliberately estimating.

Worked Example 5**GDC USEFUL**

A radius is 2.34 cm. Evaluate πr^2 and report to 3 significant figures.

WORKING / ANNOTATIONS

Checkpoint - 1.1C

1. Round 0.006953 to 2 s.f. and write the answer so the precision is visible.
2. How many significant figures are in 3.0400?
3. Explain why 84.000000 on a calculator does not prove six decimal places of accuracy.
4. Distinguish an approximation from an estimate in one sentence.

WORKING / ANNOTATIONS

1.1D - Bounds and Error

intervals, endpoint choices and percentage error

Direct error interval

If x_0 is rounded to the nearest unit h ,

$$x_0 - \frac{h}{2} \leq x < x_0 + \frac{h}{2}.$$

For 8.4 to the nearest 0.1,

$$8.35 \leq x < 8.45.$$

The lower endpoint is included and the upper endpoint is excluded under the usual half-up convention.

Worked Example 6

GDC NOT NEEDED

State the interval for 72 kg to the nearest kilogram and for 3.8 s to the nearest 0.1 s.

WORKING / ANNOTATIONS

Your Turn 6

GDC NOT NEEDED

Write each error interval.

- (a) 260 m to nearest 10 m.
- (b) 0.047 to 2 s.f.
- (c) 8.20 kg to 3 s.f.

WORKING / ANNOTATIONS

Calculated bounds for positive quantities

Sum lower = $x_L + y_L$, upper endpoint = $x_U + y_U$.

Difference lower = $x_L - y_U$, upper endpoint = $x_U - y_L$.

Product lower = $x_L y_L$, upper endpoint = $x_U y_U$.

Power for increasing positive x^n , use x_L^n and x_U^n .

Quotient lower = x_L / y_U , upper endpoint = x_U / y_L .

If intervals cross zero or the expression is not monotonic, analyze the possible extremes rather than applying a memorized positive rule.

Worked Example 7

GDC NOT NEEDED

A rectangle is measured as 12.4 m by 7.2 m, each to the nearest 0.1 m. Find area bounds.

WORKING / ANNOTATIONS

Worked Example 8

GDC NOT NEEDED

A distance is 150 m to nearest 10 m and a time is 12 s to nearest second. Find possible speed bounds.

WORKING / ANNOTATIONS

Your Turn 7

GDC USEFUL

- (a) A square side is 8.3 cm nearest 0.1 cm. Find the upper area endpoint.
- (b) Mass 250 g nearest 10 g; volume 80 cm³ nearest 5 cm³. Find the lower density bound.
- (c) Radius 4.2 cm and height 11.5 cm, each nearest 0.1 cm. Find the upper cylinder-volume endpoint.

WORKING / ANNOTATIONS

Absolute, relative and percentage error

Let A be an approximation and X the exact/accepted reference value.

$$E_a = |A - X|, \quad E_r = \frac{|A - X|}{|X|}, \quad E\% = \frac{|A - X|}{|X|} \times 100\%.$$

Absolute error retains the original units; relative and percentage error compare the error with the reference magnitude.

Worked Example 9

GDC NOT NEEDED

An accepted value is 50 and an approximation is 48.6. Find the absolute and percentage error.

WORKING / ANNOTATIONS

Worked Example 10

GDC USEFUL

A length is reported as 8.4 cm to the nearest 0.1 cm. Find the maximum possible percentage error relative to the true value.

WORKING / ANNOTATIONS

Checkpoint - 1.1D

1. State the interval for 4.60 to 3 s.f.
2. State the maximum absolute rounding error when rounding to nearest 0.01.
3. If $2 \leq x < 3$ and $5 \leq y < 7$, state bounds for $x + y$.
4. For positive x, y , state the lower quotient rule.
5. An accepted value is 200 and an approximation is 196. Find absolute and percentage error.

WORKING / ANNOTATIONS

1.1E - Integrated Modelling and Assessment

combine scale, precision, bounds, technology and interpretation

IB-style task 1 - Solar data archive

GDC EXPECTED

A satellite stores 3.6×10^5 images. Each image requires 2.4×10^6 bytes.

- Calculate the total storage in bytes, in standard form. [2]
- A compression system reduces storage by 37%. Find the compressed storage. [2]
- The archive is copied at 1.5×10^8 bytes per second. Find the transfer time in minutes. [3]
- State one reason the calculated transfer time may differ from reality. [1]

WORKING / ANNOTATIONS

IB-style task 2 - Water-use estimation

GDC EXPECTED

A campus has 1450 people. A preliminary model assumes each person uses 1.8×10^2 L of water per day on 210 operating days each year.

- Estimate annual water use in litres and write it in standard form. [3]
- Convert the result to cubic metres, using $1 \text{ m}^3 = 10^3 \text{ L}$. [2]
- A conservation plan targets a 12% reduction. Calculate the target annual use in cubic metres. [2]
- Give two assumptions that should be tested before budgeting from the model. [2]
- Explain why reporting the target to the nearest litre would be misleading. [1]

WORKING / ANNOTATIONS

IB-style task 3 - Precision audit

GDC USEFUL

A circular optical component has measured diameter $d = 4.20$ cm to 3 significant figures. The model is $A = \pi d^2/4$.

- Write the lower and upper endpoints for d . [2]
- Calculate lower and upper endpoints for the area. [3]
- The nominal model gives 13.85 cm². Determine a suitable upper bound for the percentage error relative to any true area consistent with the measurement. [3]
- Comment on whether reporting the area as 13.854423 cm² is appropriate. [2]

WORKING / ANNOTATIONS

Mastery checklist

Before leaving Lesson 1.1, confirm that you can:

- normalize and expand scientific notation without losing signs or powers;
- align exponents before addition or subtraction;
- convert squared/cubed units using squared/cubed conversion factors;
- estimate before trusting calculator output;
- use EE, SCI/NORM, the negation key, brackets, and guard digits correctly;
- show precision through zeros and justified rounding;
- construct direct and calculated bounds with correct endpoint choices;
- calculate and interpret absolute/relative/percentage error;
- state why a model or measurement does not justify excessive precision.

Checkpoint - Final core mastery

1. Calculate $\frac{5.4 \times 10^9}{9.0 \times 10^2}$ and give a one-line scale check.
2. Round 0.004986 to 3 s.f.
3. A value is reported as 0.047 to 2 s.f. State its error interval.
4. Two dimensions are positive and rounded. Explain which endpoints produce an upper product bound.
5. An approximation differs from an accepted value by 1.2%. State what additional information is needed before deciding whether the error is acceptable.

WORKING / ANNOTATIONS

Optional Extension Appendix

prerequisite review and alternate scale convention

OPTIONAL EXTENSION Number-set prerequisite review

The live Lesson 1.1 retains number-set material for revision, but it is excluded from default core mastery. The hierarchy is

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

This lesson uses the convention $0 \notin \mathbb{N}$.

Quick classification reference

$\mathbb{N}$	counting numbers $1, 2, 3, \dots$
$\mathbb{Z}$	integers $\dots, -2, -1, 0, 1, 2, \dots$
$\mathbb{Q}$	fractions of integers; terminating or recurring decimals
$\mathbb{R} \setminus \mathbb{Q}$	irrational real numbers such as $\sqrt{2}$ and π
$\mathbb{C}$	complex numbers $a + bi$, where $i^2 = -1$

Your Turn Extension 1

OPTIONAL EXTENSION

State the smallest appropriate set for -29 , 0.042 , $\sqrt{50}$, $0.\overline{405}$, 11 , and $-4i$.

WORKING / ANNOTATIONS

OPTIONAL EXTENSION Nearest-power order of magnitude

The core route compares scale using exponents, estimates, and ratios without requiring logarithms. If a question explicitly defines a *nearest power of ten* convention, the boundary between 10^k and 10^{k+1} is $\sqrt{10} \times 10^k$.

$$N = a \times 10^k, \quad 1 \leq a < 10 : \quad \begin{cases} \text{nearest power } 10^k, & a < \sqrt{10}, \\ \text{nearest power } 10^{k+1}, & a > \sqrt{10}. \end{cases}$$

Use this only when the question defines or requires that convention.

Final perspective

Scientific notation is not just a formatting rule. It links **scale, operations, units, calculator literacy, precision, and uncertainty**. The complete mathematical habit is:

Estimate first → calculate carefully → report honestly → interpret in context.